

Data_Scientist — Step 6

Learn Data Visualization

Detailed Learning Notes • CareerCompass

1. Learning Objective

Communicate analytical findings with accurate, readable and decision-oriented visualizations.

2. Core Concepts — Detailed Breakdown

Chart selection

Match chart type to question: bars for category comparisons, lines for time trends, scatter plots for relationships, histograms/box plots for distributions and maps only when geography is analytically relevant.

Visual hierarchy

Use position and aligned scales where possible. Reduce clutter, label units, use meaningful titles and annotate important points.

Exploratory vs explanatory

EDA charts help you investigate; presentation charts should emphasize a clear conclusion and audience need.

Dashboards

A dashboard should have a logical reading order, filters with clear purpose and consistent definitions. Avoid filling every space with a chart.

Ethical visualization

Avoid misleading scales, cherry-picked windows, unnecessary 3D effects and unexplained aggregations.

3. Recommended Learning Workflow

- Start from the business/research question.
- Build a simple chart.
- Check axes and aggregation.
- Add labels and context.
- Ask a colleague to interpret it without explanation.
- Refine based on whether the intended message is understood.

5. Hands-on Project / Practice

Create a dashboard/report from a public dataset with an executive summary, KPI section, trend analysis, category comparison, distribution and one deeper diagnostic view. Write a short design rationale explaining every major chart.

6. Common Mistakes & Pitfalls

- Overloading dashboards.
- Using pie charts for many categories.
- Using color without accessibility considerations.

- Confusing correlation with causation.
- Showing metrics without defining the denominator or time window.

7. Completion Checklist

- I can choose appropriate charts.
- I can explain every chart's analytical purpose.
- I can design a readable dashboard.
- I can identify misleading visualizations.
- I can communicate findings to a non-technical audience.

8. Interview / Assessment Questions

- Why is a bar chart often better than a map for category comparison?
- What makes a dashboard effective?
- How can aggregation mislead?
- What is the difference between exploratory and explanatory visualization?
- How do you design for accessibility?

9. Recommended References

Tableau Free Training: <https://www.tableau.com/learn/training>

Tableau visualization guidance: <https://www.tableau.com/learn/get-started/desktop-viz-design>

pandas User Guide: https://pandas.pydata.org/docs/user_guide/

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.